

I GUSTI BAGUS RAMADHA SAVERIAN RANUH

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EDUCATION

BINUS UNIVERSITY – Jakarta, Indonesia
Bachelor of Computer Science - GPA 3.94/4.00

September 2022 – March 2026

EXPERIENCE

AI Engineer Intern at GDP Labs – Jakarta, Indonesia

February 2025 – February 2026

Internship at GDP Labs

- Optimized LLM inference to support high-concurrency user workloads, achieving a **1.8× increase** in throughput and up to **70% faster latency** by migrating vLLM to the SGLang framework.
- Integrates a new relevancy & correctness metric in the internal evaluation SDK, expanded coverage to approximately **30% of total production** queries that were previously not reliably evaluable.
- Streamline deployment workflows by containerizing inference services with Docker Compose, **enabling one-line deployment** for inference and evaluation.

Data Science Dicoding Bootcamp – Online

October 2024 – February 2025

Bootcamp at Dicoding

- Complete **600+ hours** of an intensive data science and ML program for 6 months, designed to equip digital talents with industry-standard skills.
- Develop multiple hands-on projects with final capstone projects that apply data science and machine learning techniques to real-world problems.

Freshmen Partner – Semarang, Indonesia

September 2023 – July 2024

Freshman Partner at BINUS University

- Guides **10 freshmen** by sharing essential experiences and materials to help them navigate first-year university life with focus and enjoyment.
- Facilitate weekly discussion and sharing sessions, fostering an inclusive environment that encourages open communication, peer support, and collaborative problem-solving.

Tutor – Semarang, Indonesia

September 2023 – January 2024

Tutor at BINUS University

- Conducts weekly study classes for Algorithm & Programming and Discrete Math to first-semester students throughout one semester, consistently attracting over **10-20 students**.
- Facilitates sessions to assist students struggling with course material, helping them improve their understanding and achieve their academic goals in the subjects.

Applied Mathematics Exchange Student at CYCU – Online

February 2023 – July 2023

International Exchange Program at Chung Yuan Christian University

- Studying foundational mathematical models, applied mathematics, and statistics; strengthening quantitative analysis and critical thinking.
- Gain **exposure to an international academic** environment early in my academic career.

ACHIEVEMENTS

Researcher at ISRITI – Yogyakarta, Indonesia

December 2025

Researcher at ISRITI held by IEEE

- Conduct a research paper presented at the International Seminar on Research of Information Technology and Intelligent System (ISRITI) with a Scopus index, focusing on developing an autonomous multi-agent orchestration framework designed to manage the end-to-end lifecycle of synthetic data generation, from raw text retrieval and processing to validation and annotation.

Researcher at ICCSCI – Jakarta, Indonesia

August 2024

Researcher at ICCSCI held by Procedia Computer Science and Elsevier

- Conduct a research paper presented at the International Conference on Computer Science and Computational Intelligence (ICCSCI) with a Scopus index, focusing on a comparative analysis of deep learning algorithms for the classification of hyperpigmented skin diseases. The study explored performance metrics across various models to optimize diagnostic accuracy.

10th National Finalist ADSE Competition – Jakarta, Indonesia

April 2024 – August 2024

National Finalist ASEAN Data Scientist Explorers Competition Issued by ASEAN Foundation and SAP

- Develop a pioneering data-driven application and solution for the 2024 ASEAN Data Scientist Explorers Competition, which focuses on “Enhancing Waste Management through Integrated Waste Banks and Gamification Incentives,” and aims to advance sustainable waste management practices across ASEAN countries.

10th Finalist SOCS AI For Accessibility Hackathon – Jakarta, Indonesia

April 2024

Finalist School of Computer Science Hackathon 2024 BINUS x Microsoft (AI4A)

- Builds a mobile web application designed to assist those with vision impairments in identifying objects and text from a distance or from areas inaccessible with a white cane (upper body). making use of Microsoft Azure features, including Speech Service, Translator, and Computer Vision. Everything will be spoken using text-to-speech.

SKILLS

- **Programming Language:** Python, JavaScript, C++
- **Frameworks:** SGLang, vLLM, HuggingFace, LangChain, Streamlit, DeepEvals
- **Database Management System:** MySQL, MongoDB, Supabase
- **Data Visualization:** Tableau, SAP Analytics Cloud
- **Deep Learning:** PyTorch, TensorFlow
- **DevOps:** Git/GitHub, Docker, Docker Compose
- **Data Science & Miscellaneous Technology:** A/B Testing, ETL, Data Pipeline, Statistics, Time series, Experimental design, Hypothesis testing, OOP, APIs
- **Cloud:** GCP (BigQuery, Cloud Storage, Vertex AI Studio), Azure (Speech, Vision, Translator)
- **Natural Language Processing (NLP):** LLM, RAG, Embedding, Multimodality, Agentic
- **Computer Vision (CV):** Image Classification, OCR, Transfer Learning with CNN (ResNet, MobileNet), Real-time image inference

PROJECTS

UMKM Fashion Sales Dashboard

Python, PCA, Kmeans, SVM, SARIMAX, SVD

Develop an end-to-end UMKM Fashion Sales Dashboard to support data-driven decision-making for small and medium fashion businesses by enabling real-time sales monitoring, customer behavior analysis, demand forecasting, and a recommendation system. The solution features RFM-based customer segmentation with PCA and K-Means clustering, classification models (SVM, Logistic Regression, Naive Bayes, Decision Tree), where SVM achieved the highest F1-score due to its effectiveness in handling high-dimensional and non-linear decision boundaries, time-series forecasting using SARIMAX for future sales prediction, and an SVD model for item recommendations.

SafeSight: Object Detection and Text Recognition for the Visually Impaired

Bootstrap, JavaScript, Node.js, Express.js, Docker, Cloudflare, Microsoft Azure

A mobile web application designed to assist visually impaired person in detecting objects and recognizing text in areas beyond their reach, where a white cane cannot be used for the upper body. It uses Microsoft Azure services, including Computer Vision, Translator, and Speech Services, to provide real-time verbal feedback through text-to-speech, ensuring seamless accessibility and navigation.

Optimize LLM Inference using SGLang

Python, SGLang, CUDA, TensorFlow, Keras, Librosa

Evaluate and optimize LLM inference performance using SGLang as the primary inference server, conducting a comparative study against vLLM under high concurrent workloads. The scope encompasses systematic experimentation and hyperparameter tuning across various optimization strategies, including quantization, data parallelism, and tensor parallelism. The resulting analysis provided actionable insights on production trade-offs between throughput, latency, memory efficiency, and model fidelity, enabling multiple internal AI teams to make informed deployment decisions.

NLP Multimodal RAG-Based Chatbot Model

Python, LLM, RAG, Llama CCP, Llama Index, Gradio

Implement an AI chatbot utilizing Large Language Models (LLMs) and Retrieval Augmented Generation (RAG) techniques to provide detailed answers based on knowledge extracted from PDF documents. This implementation integrates LlamaCPP for LLM functionality and llama_index for effective document management. The system leverages medical knowledge from PDF documents to provide fine-tuned response.

Deep Learning Algorithms for Image-Based Classification of Hyperpigmented Skin Disease

Python, YOLO, DenseNet201, GoogleNet, InceptionResNetV2, MobileNet

Analyze and compare hyperpigmented skin disease using machine learning and deep learning techniques. The study analyzed pretrained models like YOLO, DenseNet201, GoogLeNet, InceptionResNetV2, and MobileNet. The study highlights DenseNet201 as the best-performing model for the accurate classification of hyperpigmented skin conditions, based on both accuracy and AUC. However, YOLO was ultimately chosen due to its effective object detection capability.

ORGANIZATION

Data Science Club – Jakarta, Indonesia

September 2024 – July 2025

Member at BINUS University

- Actively collaborate with fellow members, participate in workshops, and engage in discussions to enhance skills in machine learning, data analysis, and research methodologies.
- Lead technical discussions on emerging AI trends, breaking down complex research papers into actionable insights for fellow members.

UREEKA BINUS – Jakarta, Indonesia

April 2024 – April 2025

Member at UREEKA BINUS University

- Engages in data mining training through UREEKA, collaborating in teams to compete in various data competitions.
- Develops end-to-end data pipelines for competition submissions using classic ML solutions, utilizing feature engineering and hyperparameter tuning to improve model performance metrics.

CERTIFICATES

Oral Presentation | International Seminar on Research of Information Technology and Intelligent Systems | 2025

IBM RAG and Agentic AI Professional Certificate | IBM | 2025

Dicoding Data Science Bootcamp 2025 | Dicoding Indonesia | 2025

ASEAN Data Science Explorers 2024 National Final Indonesia | ASEAN Foundation | 2024

Oral Presentation | International Conference on Computer Science and Computational Intelligence (ISSCSI) | 2024